Project 3 — BJT Emitter Follower

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1 Given Information and Design Goal

For the design:

- Source resistance: $R_S = 100\,\Omega$
- Load resistance: $R_L = 100\,\Omega$
- Input amplitude: $0.5\,\mathrm{V}$
- Thermal voltage: $V_T \approx 25\,\mathrm{mV}$

The two main requirements are:

1. Attenuation requirement: overall gain should satisfy

$$G \geq 0.9$$

2. Signal swing requirement: small-signal variation in v_{be} should remain small

The chosen topology is a common-collector (emitter follower) BJT amplifier.

2 Circuit Schematic

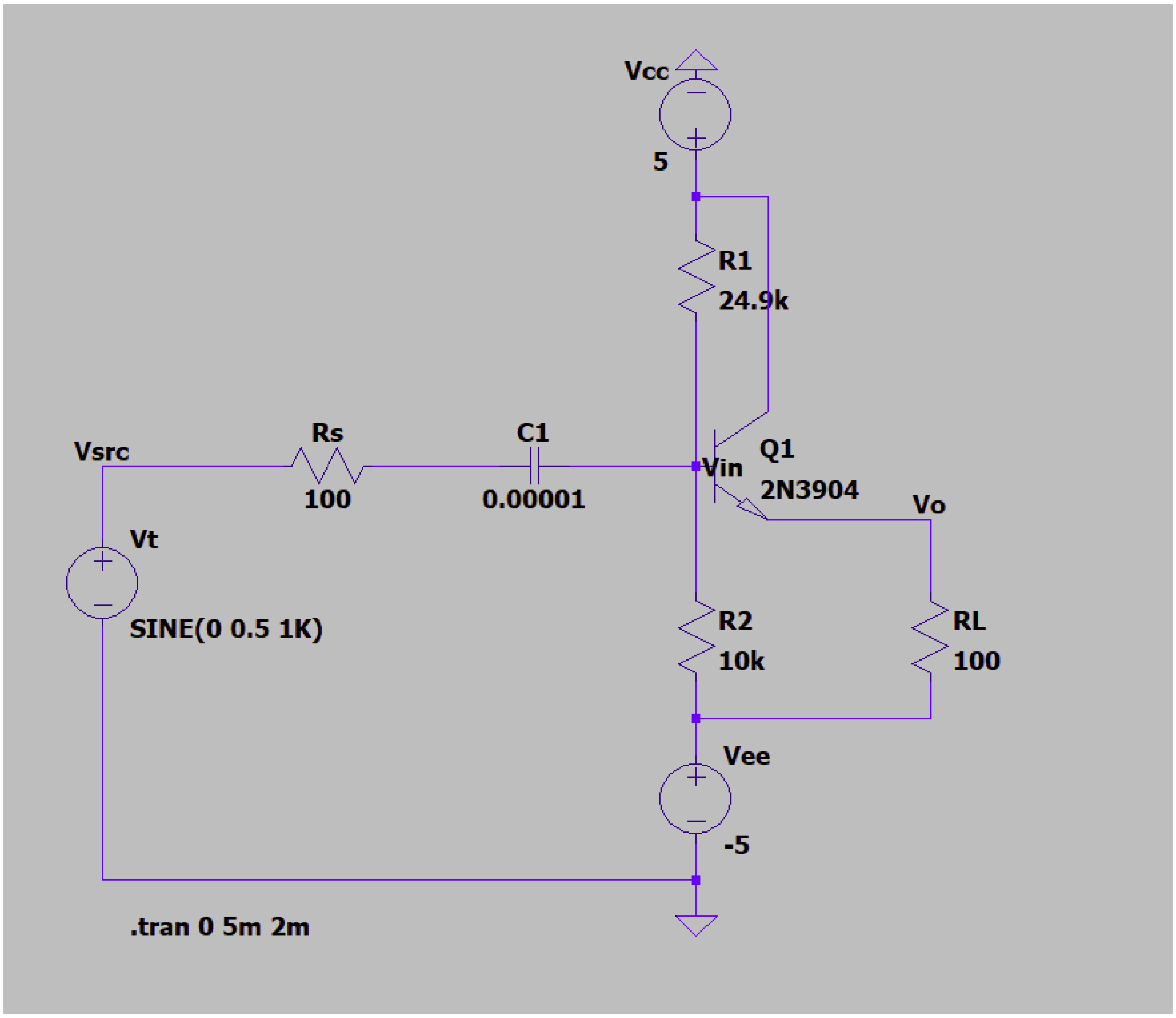


Figure 1: Simulated Circuit with SPICE

2.1 Choice of transistor

This design problem requires good buffering in order to satisfy the attenuation constraint. Amplifiers behave as sources with output resistance (R_{out}) in series with a load. Since the voltage drop across the output resistance creates a voltage divider, minimizing R_{out} is necessary to maximize the transfer function $V_{\text{out}}/V_{\text{in}}$.

A configuration that achieves this is the common-collector BJT. For this configuration, biasing the BJT in forward-active mode results in $R_{\text{out}} \approx 1/g_m$. Hence, a large g_m results in a small R_{out}. Since BJTs typically have a small V_t value ($\approx 25\,\text{mV}$) and $g_m = I_C/V_t$, achieving a low output impedance is trivial. Minimizing R_{out} in the MOSFET case is more challenging because $R_{\text{out}} = 1/(\lambda I_D)$ and $\lambda \ll 1$. Thus, a very large I_D value is needed to achieve a low output impedance, promoting a more inefficient design. For these reasons, the proposed solution prefers the common-collector BJT over the MOSFET to meet the design goals of $< 10\%$ signal attenuation.

2.2 Amplifier topology

As mentioned earlier, the common collector BJT was chosen because biasing them in forward active mode results in $R_{\text{out}} \approx 1/g_m$. This means that R_{out} is dominated by $1/g_m$ which is small. The other topologies do not make it as simple to minimize R_{out}. The common emitter has $R_{\text{out}} \approx r_o \approx V_A/I_C$. V_A is typically very large relative to I_C, leading to a high output impedance. The common base has $R_{\text{out}} \approx r_o(1 + g_m R_s)$. This means that R_{out} is directly proportional to g_m (unlike the common collector case) so the large g_m value (due to $g_m = I_C/V_t$ and V_t being small), leads to $g_m R_s \gg 1$ which forces R_{out} to be large. Therefore, the common base and common emitter topologies are not ideal for this design problem.

2.3 Small-Signal Model

The small-signal emitter resistance is

$$r_e = \frac{1}{g_m}$$

Using

$$g_m = \frac{I_C}{V_T}$$

gives

$$r_e = \frac{V_T}{I_C}$$

Since for a BJT in active region,

$$I_E \approx I_C$$

we use

$$r_e \approx \frac{V_T}{I_E} \approx \frac{25\,\text{mV}}{I_E}$$

Also, the input resistance is approximated by

$$R_{in} = R_1 \parallel R_2 \parallel (\beta + 1)(r_e + R_L)$$

2.4 Attenuation Requirement

The overall gain is approximated by

$$G = \frac{R_{in}}{R_{in} + R_S} \cdot \frac{R_L}{r_e + R_L}$$

The design requirement is

$$G \geq 0.9$$

Substituting the known values $R_S = 100\,\Omega$ and $R_L = 100\,\Omega$:

$$\frac{R_{in}}{R_{in} + 100} \cdot \frac{100}{r_e + 100} \geq 0.9$$

To satisfy this expression, we want:

$$R_{in} \gg 100 \quad \text{and} \quad r_e \ll 100$$

As a first design target, choose

$$R_{in} \approx 10\,\mathrm{k}\Omega \quad \text{and} \quad r_e \approx 1\,\Omega$$

Then

$$G \approx \frac{10000}{10100} \cdot \frac{100}{101}$$
$$G \approx 0.9901 \cdot 0.9901$$
$$G \approx 0.98$$

Therefore,

$$G \approx 0.98 > 0.9$$

so the attenuation requirement is satisfied.

2.5 Signal Swing Requirement

The signal swing requirement was written as

$$0.5 \cdot \frac{R_{in}}{R_{in} + R_S} \cdot \frac{r_e}{r_e + R_L} \leq 0.2x$$

where

$$x = 25\,\mathrm{mV}$$

So the right-hand side becomes

$$0.2x = 0.2(25\,\mathrm{mV}) = 5\,\mathrm{mV}$$

Thus the condition is

$$0.5 \cdot \frac{R_{in}}{R_{in} + R_S} \cdot \frac{r_e}{r_e + R_L} \leq 5\,\mathrm{mV}$$

Substituting $R_S = 100\,\Omega$ and $R_L = 100\,\Omega$ gives

$$0.5 \cdot \frac{R_{in}}{R_{in} + 100} \cdot \frac{r_e}{r_e + 100} \leq 5\,\mathrm{mV}$$

Using the same design target as before,

$$R_{in} \approx 10\,\mathrm{k}\Omega \quad \text{and} \quad r_e \approx 1\,\Omega$$

we get

$$0.5 \cdot \frac{10000}{10100} \cdot \frac{1}{101}$$
$$= 0.5 \cdot 0.9901 \cdot 0.00990$$
$$\approx 0.0049\,\mathrm{V}$$
$$\approx 4.9\,\mathrm{mV}$$

Hence,

$$4.9\,\mathrm{mV} < 5\,\mathrm{mV}$$

so the signal swing requirement is also satisfied. This confirms that choosing

$$R_{in} \approx 10\,\mathrm{k}\Omega \quad \text{and} \quad r_e \approx 1\,\Omega$$

is a safe design target.

2.6 Choosing the Bias Current from r_e

Since

$$r_e \approx \frac{25\,\text{mV}}{I_E}$$

and we want

$$r_e \approx 1\,\Omega$$

then

$$1 \approx \frac{25\,\text{mV}}{I_E}$$

so

$$I_E \approx 25\,\text{mA}$$

For the purposes of design, use

$$I_C \approx I_E \approx 25\,\text{mA}$$

2.7 DC Bias Target

The emitter resistor is $R_L = 100\,\Omega$, connected to the $-5\,\text{V}$ rail. Thus the emitter voltage is estimated as

$$V_E \approx -5 + I_E R_L$$

Substituting $I_E \approx 25\,\text{mA}$ gives

$$V_E \approx -5 + (25\,\text{mA})(100\,\Omega)$$

$$V_E \approx -5 + 2.5$$

$$V_E \approx -2.5\,\text{V}$$

Assuming the transistor is in active region,

$$V_{BE} \approx 0.7\,\text{V}$$

so

$$V_B \approx V_E + 0.7$$

$$V_B \approx -2.5 + 0.7$$

$$V_B \approx -1.8\,\text{V}$$

Therefore, the base DC bias should be around

$$V_B \approx -1.8\,\text{V}$$

2.8 Voltage Divider Design

For the DC case, the AC source is ignored and the coupling capacitor is treated as open, so the base voltage is set by the divider between $+5\,\text{V}$ and $-5\,\text{V}$. Using the divider expression:

$$V_B = V_{CC} - \frac{R_1}{R_1 + R_2}(V_{CC} - V_{EE})$$

Here,

$$V_{CC} = 5, \qquad V_{EE} = -5$$

so

$$V_{CC} - V_{EE} = 10$$

Substituting the desired base voltage $V_B \approx -1.8\,\mathrm{V}$:

$$-1.8 = 5 - \frac{R_1}{R_1 + R_2}(10)$$

Rearranging:

$$5 + 1.8 = \frac{R_1}{R_1 + R_2}(10)$$

$$6.8 = 10\frac{R_1}{R_1 + R_2}$$

$$\frac{R_1}{R_1 + R_2} = 0.68$$

This implies

$$R_1 \approx 2.125 R_2$$

So the divider should satisfy approximately

$$R_1 : R_2 \approx 2.125 : 1$$

2.9 First Trial Values

A first trial was to use

$$R_2 = 1\,\mathrm{k}\Omega, \qquad R_1 = 2.2\,\mathrm{k}\Omega$$

since this is close to the ratio found above. Using

$$r_e \approx 1\,\Omega, \qquad \beta \approx 100$$

gives

$$\begin{aligned}(\beta + 1)(r_e + R_L) &= 101(1 + 100) = 101(101)\\ &= 10201\,\Omega\end{aligned}$$

Then

$$R_{in} = 1\,\mathrm{k}\Omega \parallel 2.2\,\mathrm{k}\Omega \parallel 10201\,\Omega$$

First,

$$\begin{aligned}1\,\mathrm{k}\Omega \parallel 2.2\,\mathrm{k}\Omega &= \frac{(1000)(2200)}{1000 + 2200}\\ &= \frac{2200000}{3200} \approx 687.5\,\Omega\end{aligned}$$

Then,

$$R_{in} \approx 687.5 \parallel 10201 \approx 644\,\Omega$$

This is too small, since it is not much larger than $100\,\Omega$. That would reduce the gain too much. Therefore, this resistor choice is rejected.

2.10 Final Resistor Choice

To increase input resistance, larger divider resistors were chosen while keeping the ratio reasonably close to the target:

$$R_1 = 24.9\,\text{k}\Omega, \qquad R_2 = 10\,\text{k}\Omega$$

This gives a ratio

$$\frac{R_1}{R_2} = \frac{24.9}{10} = 2.49$$

which is slightly larger than the ideal ratio of 2.125, but still acceptable for the design. Now calculate the new input resistance:

$$R_{in} = 24.9\,\text{k}\Omega \parallel 10\,\text{k}\Omega \parallel 10201\,\Omega$$

First,

$$24.9\,\text{k}\Omega \parallel 10\,\text{k}\Omega = \frac{(24900)(10000)}{24900 + 10000}$$
$$= \frac{249000000}{34900} \approx 7135\,\Omega$$

Then,

$$R_{in} \approx 7135 \parallel 10201$$
$$R_{in} \approx 4.2\,\text{k}\Omega$$

As a rough design estimate, this is much larger than $100\,\Omega$, so it is acceptable.

2.11 Final Gain Check

Using the final design choice, the gain is estimated by

$$G = \frac{R_{in}}{R_{in} + 100} \cdot \frac{100}{101}$$

Using the rough estimate $R_{in} \approx 7.1\,\text{k}\Omega$ from the divider branch as a quick check gives

$$G \approx \frac{7.1\,\text{k}}{7.2\,\text{k}} \cdot \frac{100}{101}$$
$$G \approx 0.986 \cdot 0.990$$
$$G \approx 0.98$$

Therefore,

$$G \approx 0.98 > 0.9$$

and the attenuation requirement is satisfied.

2.12 Final Component Values

The final design values chosen were:

- $R_1 = 24.9\,\text{k}\Omega$
- $R_2 = 10\,\text{k}\Omega$
- $R_L = 100\,\Omega$
- $C_1 = 10\,\mu\text{F}$
- BJT: 2N3904
- $V_{CC} = +5\,\text{V}$, $V_{EE} = -5\,\text{V}$

3 Simulations

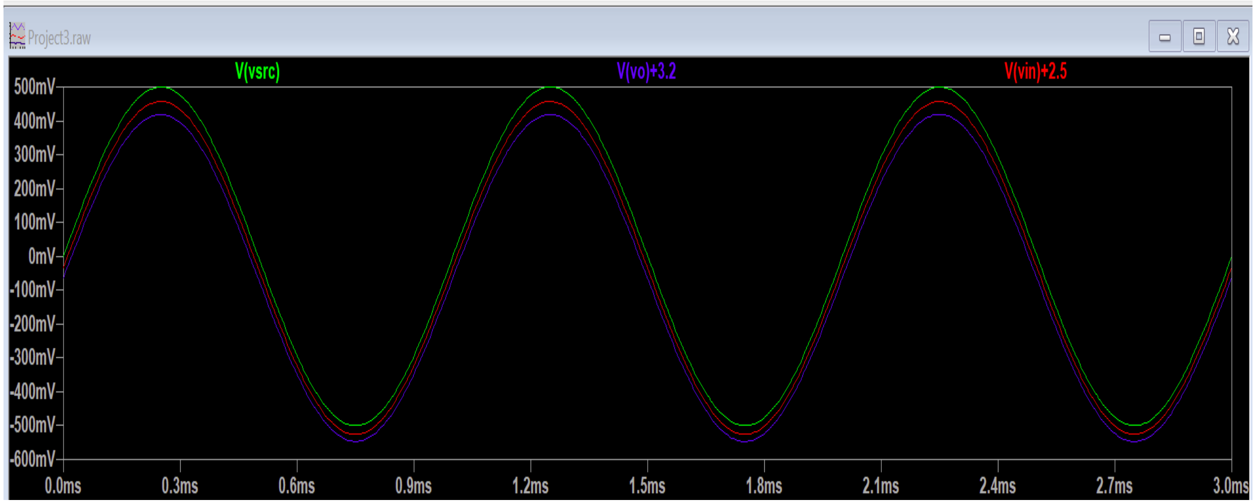


Figure 2: Simulated Circuit with SPICE

3.1 How transistor was modeled

The 2N3904 model was used in SPICE software to replicate the actual transistor with its non-ideal attributes rather than a generic transistor. This allows the simulator to capture realistic transistor behavior rather than using an idealized gain block. For theoretical analysis, the transistor was assumed to operate in forward-active mode and was treated as a small-signal common-collector.

3.2 Settings used for each simulation

A transient simulation was ran with stop time 5 ms and start time 2 ms. This allowed the startup transients to settle so the plot mainly captures the steady-state behavior. The input signal was a sine wave with 0.5 V amplitude with a 1 kHz frequency. 5 V and −5 V DC rails were used. The component values used are listed:

- $R_S = 100\,\Omega$
- $R_1 = 24.9\,\text{k}\Omega$
- $R_2 = 10\,\text{k}\Omega$
- $R_L = 100\,\Omega$
- $C_1 = 10\,\mu\text{F}$
- $Q_1 = 2\text{N}3904$

The following nodes were measured:

- V_{src} (input signal)
- V_{in} (base)
- V_{o} (emitter output)

3.3 Overall gain from simulations

```
vsrc_pp: PP(V(vsrc) )=0.999957621098 FROM 0.002 TO 0.005
vin_pp: PP(V(vin) )=0.982892513275 FROM 0.002 TO 0.005
vo_pp: PP(V(vo) )=0.965694904327 FROM 0.002 TO 0.005
g_overall: Vo_pp/Vsrc_pp=0.965735831152
a_stage: Vo_pp/Vin_pp=0.982503062425
```

Figure 3: Simulated Circuit with SPICE

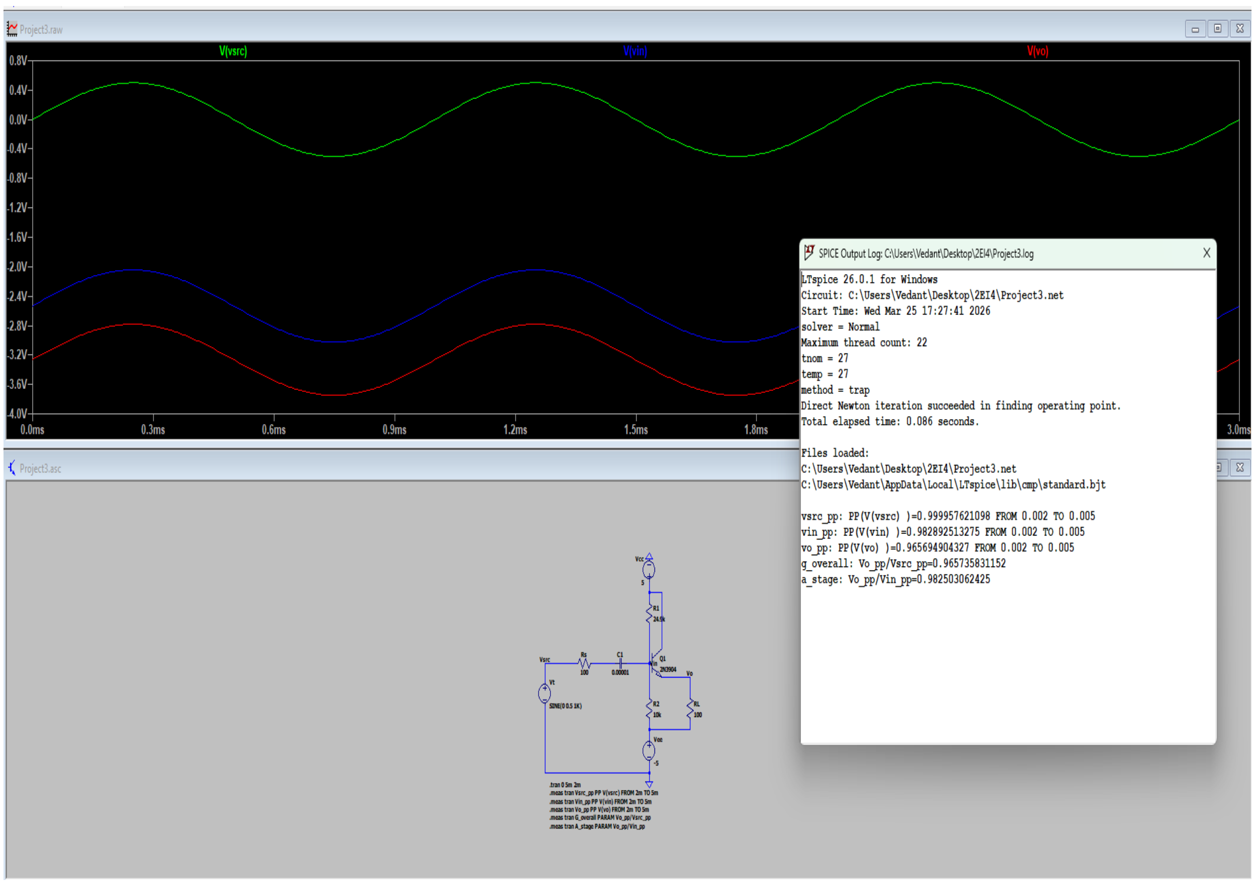


Figure 4: Simulated Circuit with SPICE

3.4 Performance parameters from simulations

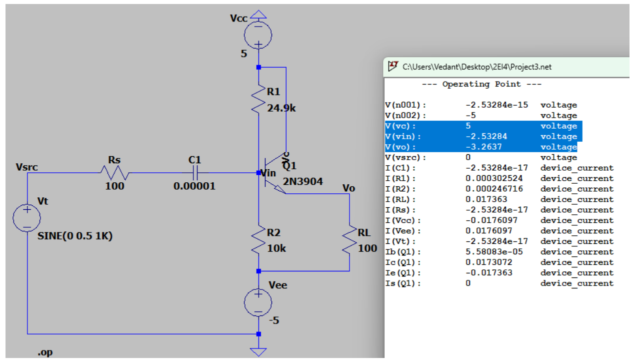
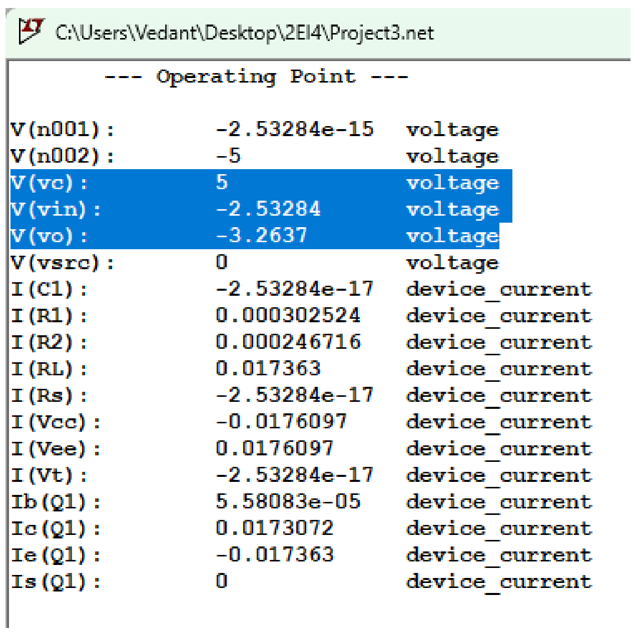


Figure 5: DC operating point


C:\Users\Vedant\Desktop\2EI4\Project3.net

--- Operating Point ---

V(n001):	-2.53284e-15	voltage
V(n002):	-5	voltage
V(vc):	5	voltage
V(vin):	-2.53284	voltage
V(vo):	-3.2637	voltage
V(vsrc):	0	voltage
I(C1):	-2.53284e-17	device_current
I(R1):	0.000302524	device_current
I(R2):	0.000246716	device_current
I(RL):	0.017363	device_current
I(Rs):	-2.53284e-17	device_current
I(Vcc):	-0.0176097	device_current
I(Vee):	0.0176097	device_current
I(Vt):	-2.53284e-17	device_current
Ib(Q1):	5.58083e-05	device_current
Ic(Q1):	0.0173072	device_current
Ie(Q1):	-0.017363	device_current
Is(Q1):	0	device_current

Figure 6: DC operating point

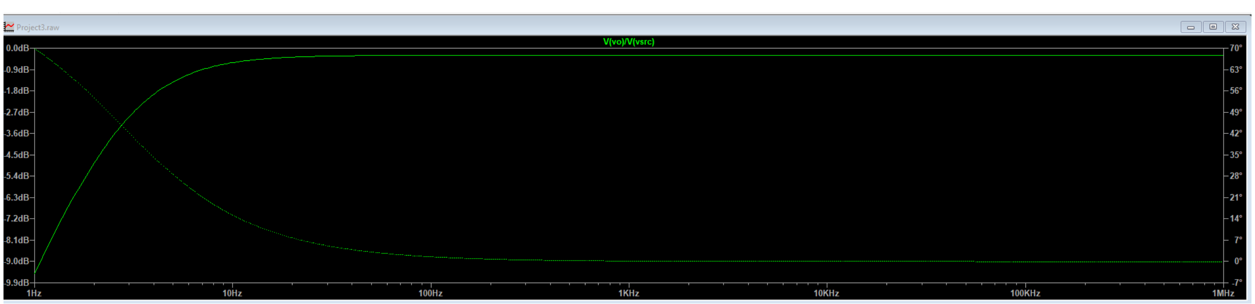


Figure 7: AC Sweep

Figures 5 and 6 use DC analysis to determine the operating point to verify that the transistor is in forward-active mode. In the screenshots, $V(\mathrm{vc})$ corresponds to the voltage at the collector, $V(\mathrm{vin})$ to the voltage at the base, and $V(\mathrm{vo})$ to the voltage at the emitter. From this, it can be deduced that $V_{be} = -2.53 - (-3.26) > 0.7\,\mathrm{V}$ and $V_{bc} = -2.53 - 5 < 0.5\,\mathrm{V}$. Therefore, the transistor is indeed forward-biased.

An AC sweep was also performed from 1 Hz to 1 MHz to determine the frequency response. The plot of V_o/V_{src} shows low-frequency attenuation due to the coupling capacitor, followed by a flat midband region. Around 1 kHz, the gain is approximately constant at about -0.30 dB, corresponding to a linear gain of about 0.966. This confirms that 1 kHz lies in the midband region and is therefore appropriate for gain measurements.

4 Physical Circuit

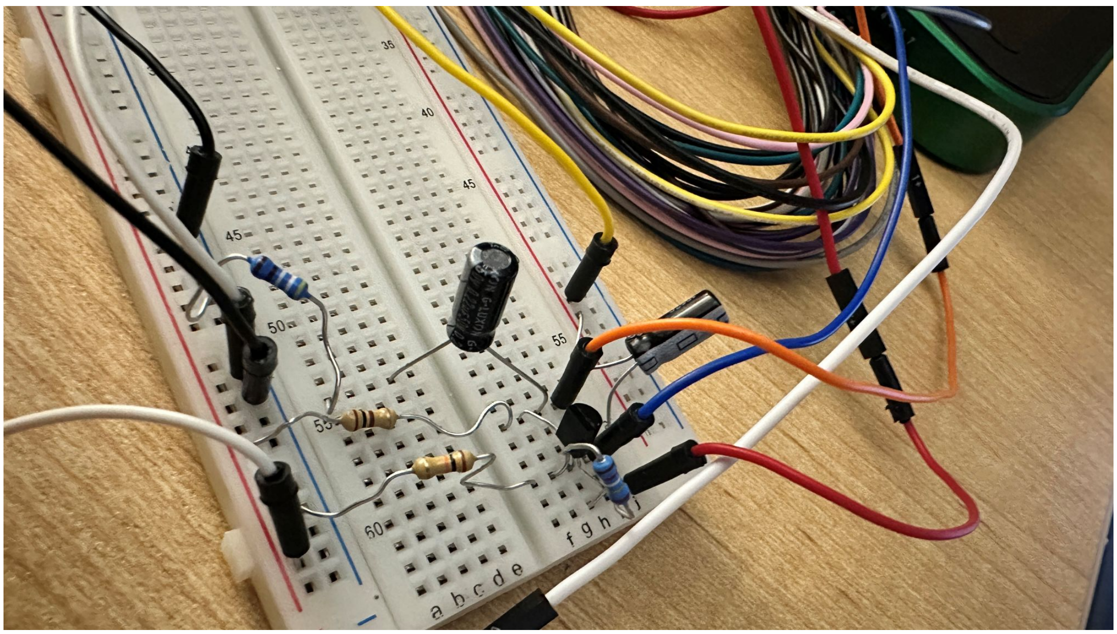


Figure 8: Physical circuit built showing breadboard and AD3

5 Measurements

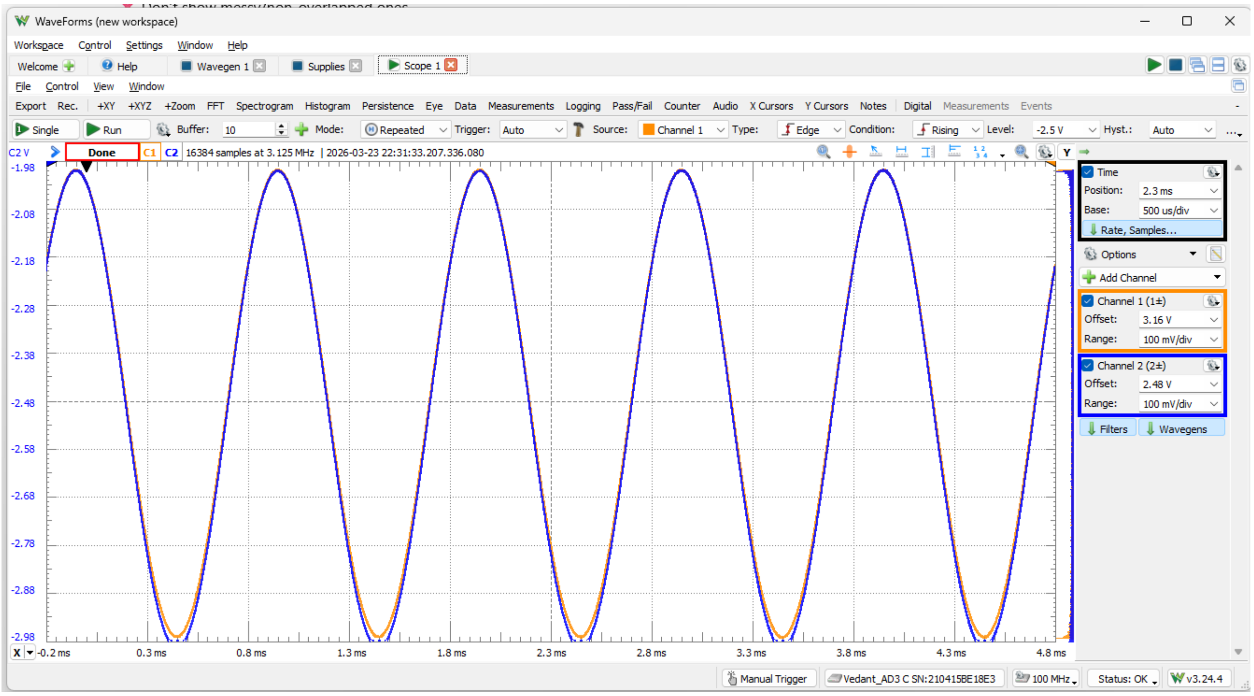


Figure 9: Waveform of output voltage vs input voltage demonstrating shape, period, and amplitude retention.

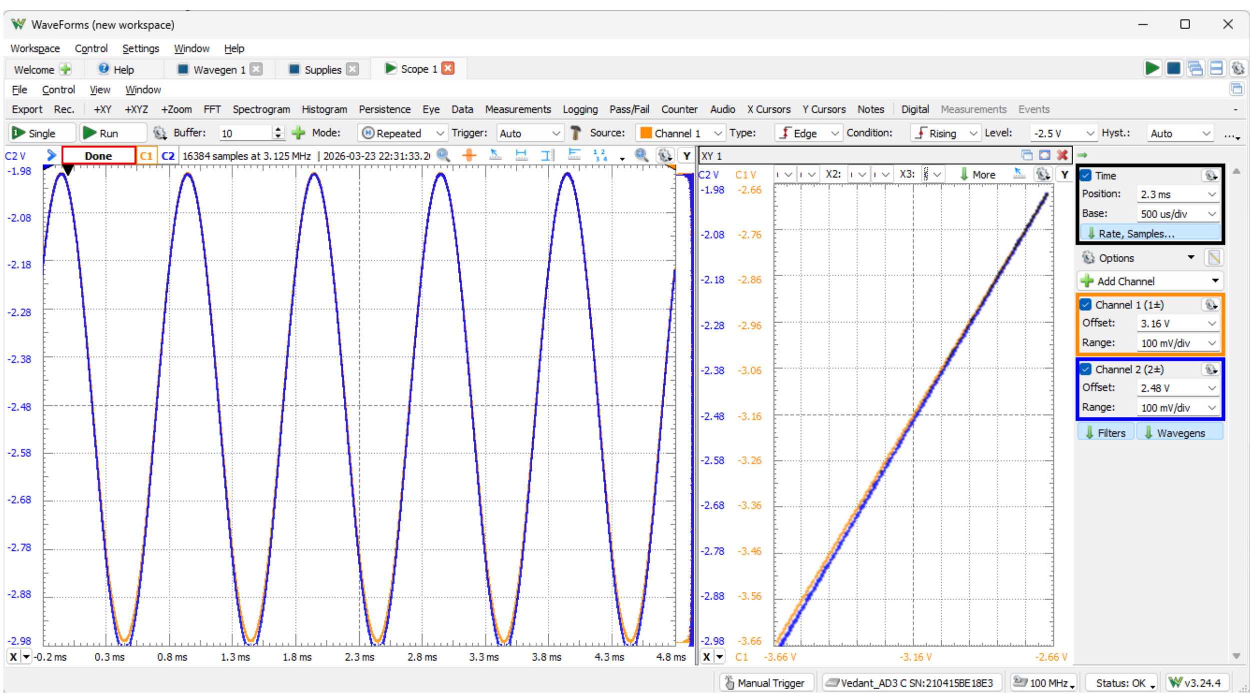


Figure 10: Waveform and XY plot of output voltage vs input voltage demonstrating linearity with a slope of approximately 1.

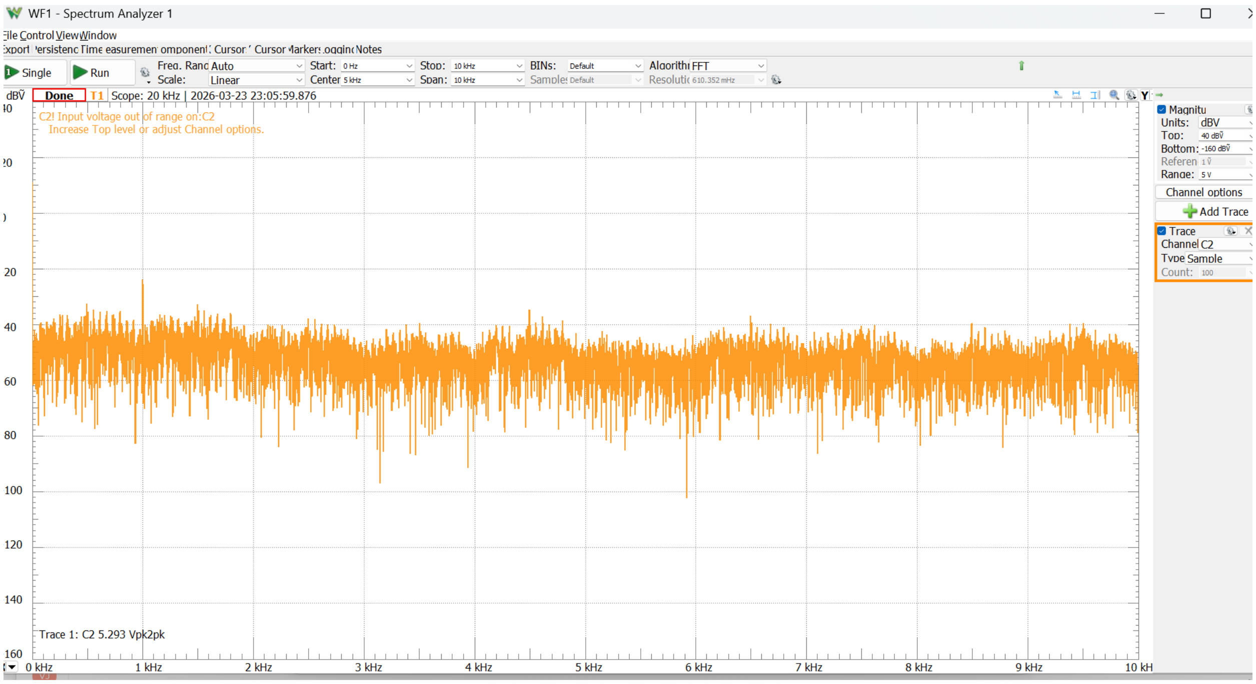


Figure 11: Spectrum analyzer

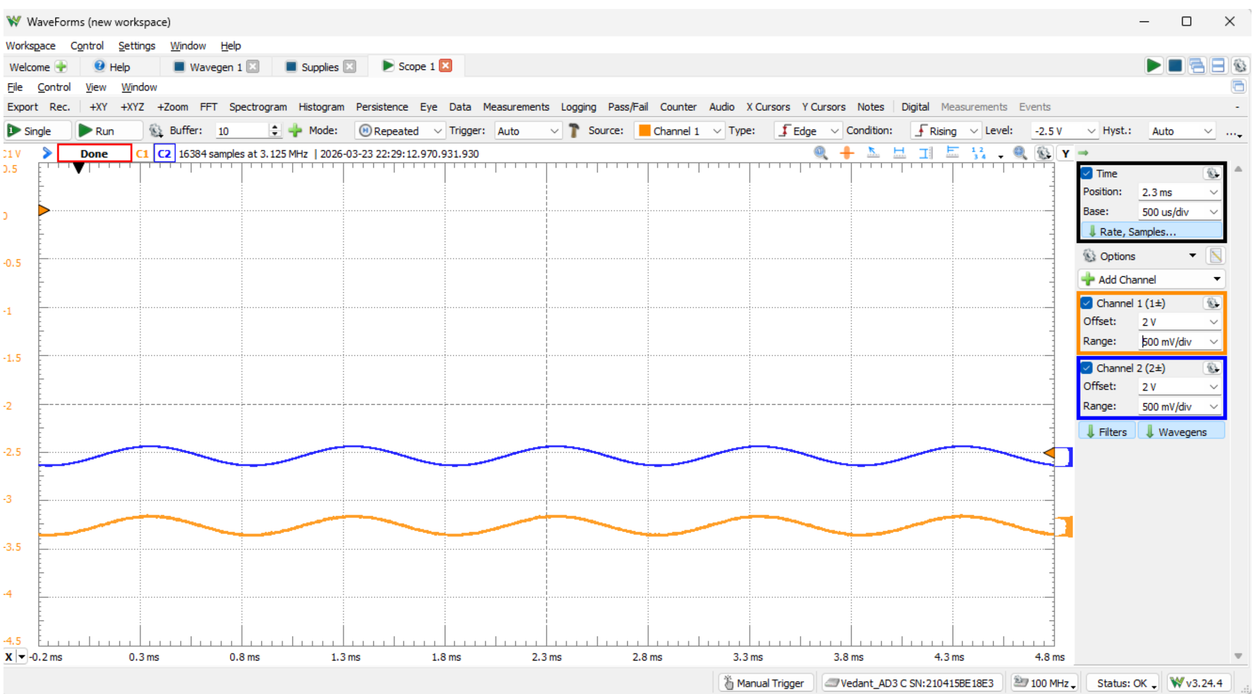


Figure 12: Measured input and output waveforms with amplitude 100 mV.

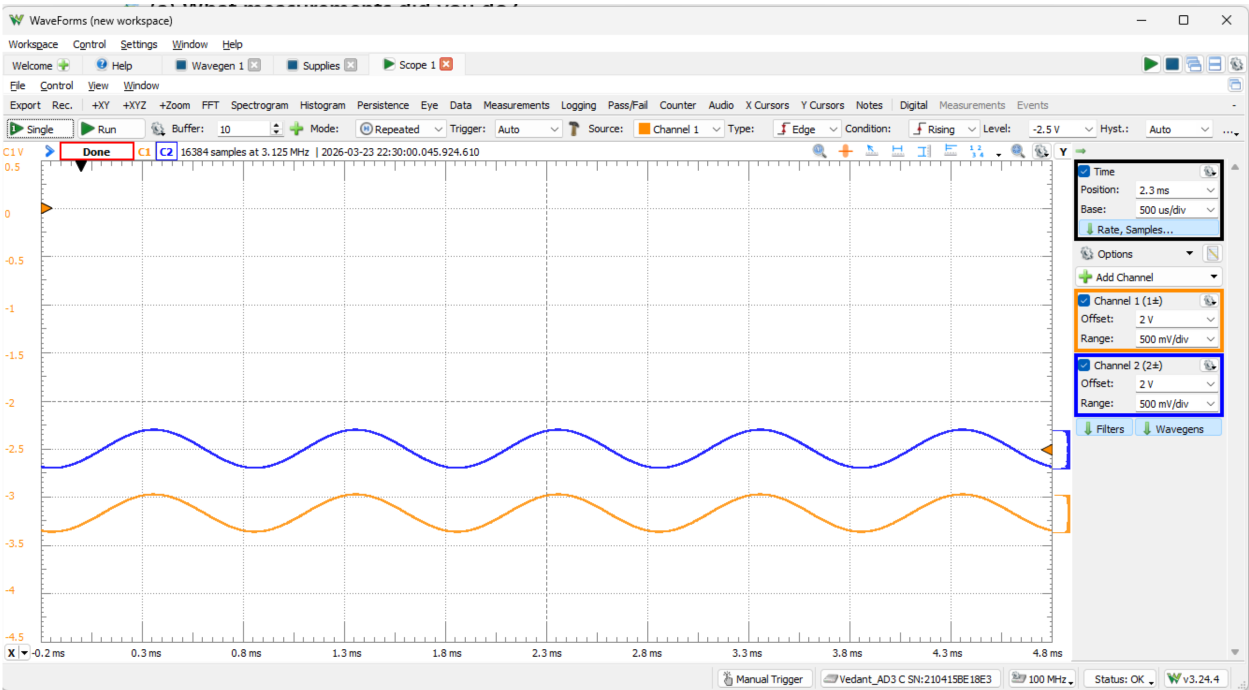


Figure 13: Measured input and output waveforms with amplitude 200 mV.

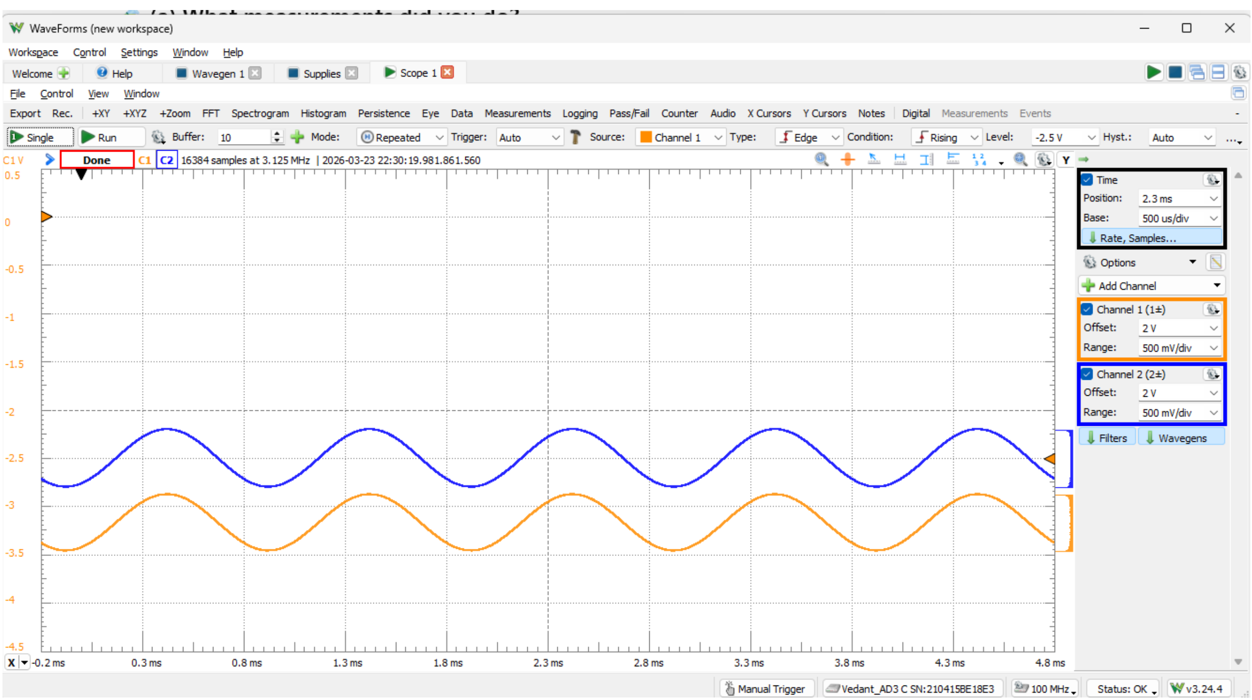


Figure 14: Measured input and output waveforms with amplitude 300 mV.

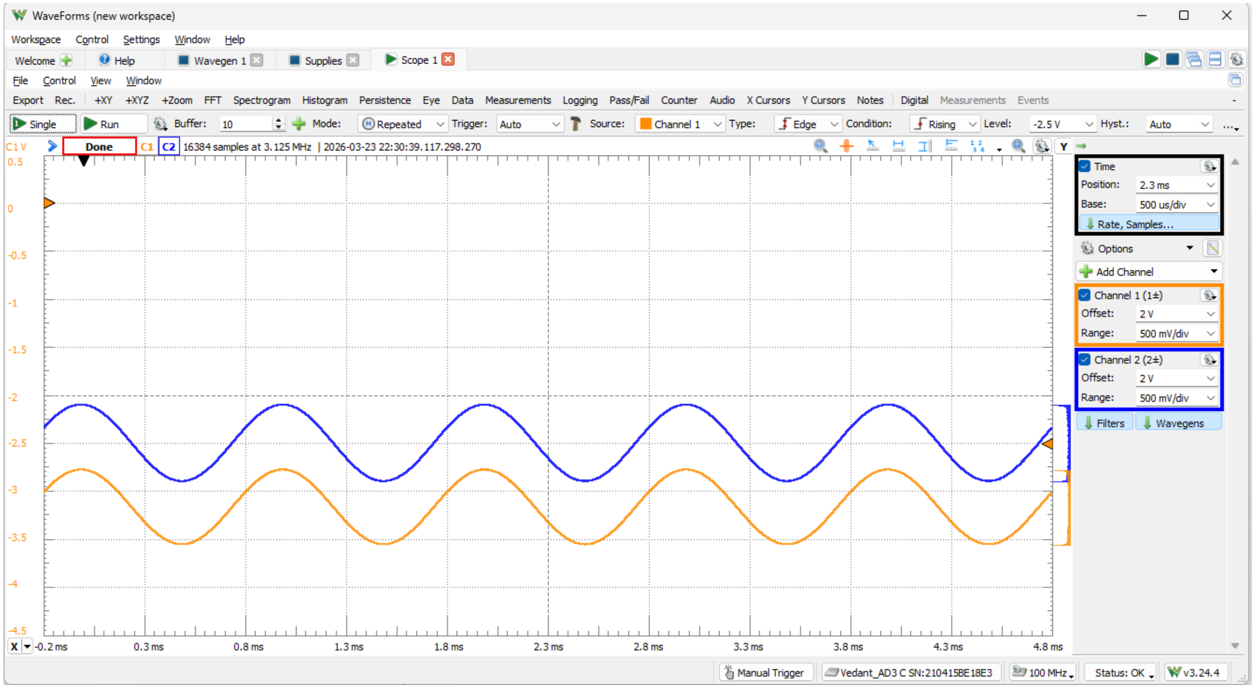


Figure 15: Measured input and output waveforms with amplitude 400 mV.

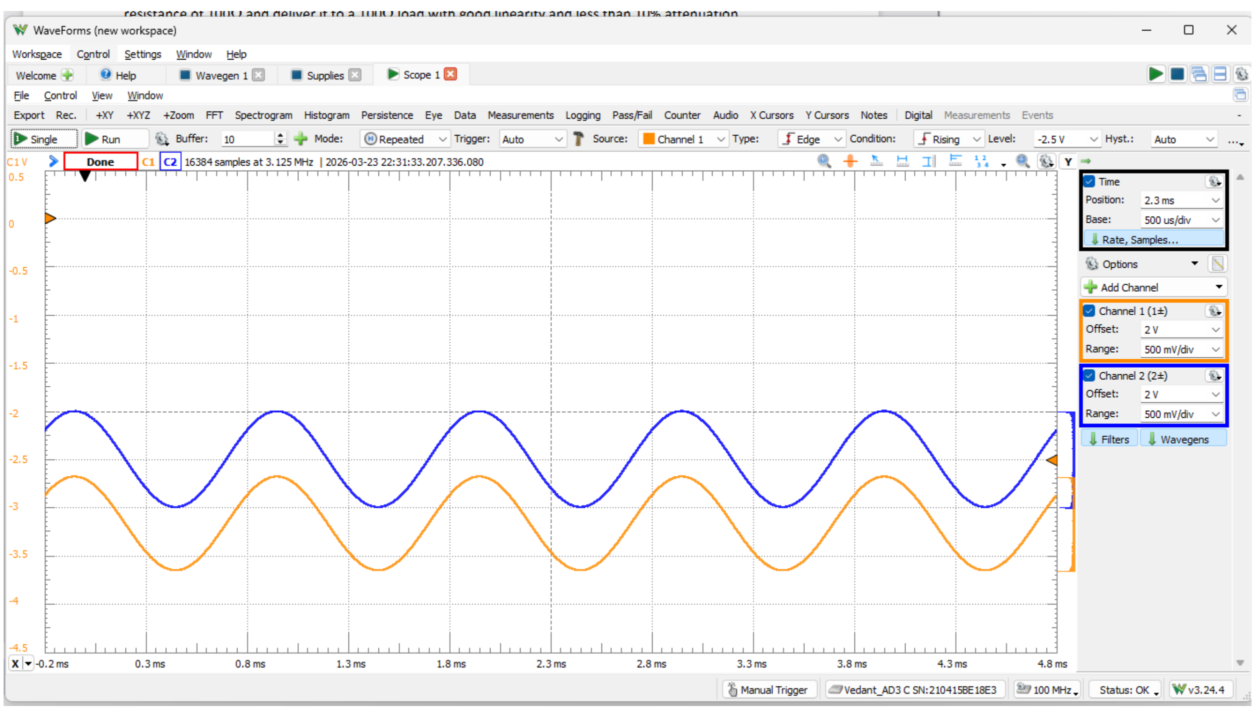


Figure 16: Measured input and output waveforms with amplitude 500 mV.

5.1 Justification of measurements

5.1.1 DC Bias measurements

The DC node voltages were measured at collector, base, and emitter to ensure the transistor was biased in the Forward-Active mode. Without this, the assumptions stated earlier would not hold. As shown in Figures 1-3, the collector was near $5\,\text{V}$, the base near $-2.5\,\text{V}$, and the emitter near $-3.2\,\text{V}$. This satisfies the Forward-Active condition of $V_{bc} = -7.5\,\text{V} < 0.5\,\text{V} = V_{bc,\text{on}}$ and $V_{be} \approx 0.7\,\text{V}$.

5.1.2 Time-domain waveform measurements

Next, the input and output waveforms were measured as functions of time. Specifically, the source signal, input signal at the base node and output signal at the emitter node were plotted as shown in Figures 4-8. Amplitude, shape, phase, and clipping were compared to verify amplitude, shape, and phase were preserved with minimal clipping. This was done before the true peak-to-peak voltages and XY plot were made as a visual demonstration of buffering. Amplitudes from $100\,\text{mV}$ to $500\,\text{mV}$ were plotted as shown.

5.1.3 Linearity measurements

From Figure 9, an XY plot comparing output vs input voltage (Figure 10) demonstrated good linearity, as it produced a straight line with a slope of ~ 1. This plot was chosen because it mathematically shows that the gain ratio is linear, as it follows a straight line. Additionally, spectrum analysis (Figure 11) was conducted as a qualitative check for harmonic distortion. Visually, that meant a lack of large spikes after the $1\,\text{kHz}$ mark as that was the frequency of the input signal. This plot was done because nonlinear distortion introduces higher-order harmonics like, in this case, $2\,\text{kHz}$ or more, and it provides a visual representation of whether the $1\,\text{kHz}$ frequency dominates or not.

5.1.4 Midband gain measurements

The peak to peak voltage was measured using the AD3 scope settings within Waveforms, and the ratio of the peak-to-peak voltages of the output signal compared to the input signal came out to be ~ 0.977 as shown in Figure 12. This was done at $1\,\text{kHz}$, a midband operating frequency.

5.2 Midband gain

To compute midband gain, a frequency of $1\,\text{kHz}$ was chosen to stay in a region outside of low-frequency rolloff (caused by the coupling capacitor(s)) and high-frequency rolloff (caused by parasitic capacitances). The peak-to-peak voltage ratio of the output signal to the input signal was computed within Waveforms software which came out to be ~ 0.977 as shown in Figure 12. This easily meets the criteria, and was approximately equal to the simulation results (with $x\%$ deviation) and calculation results (with $y\%$ deviation).

5.3 Linearity at input amplitude of $0.5\,\mathrm{V}$

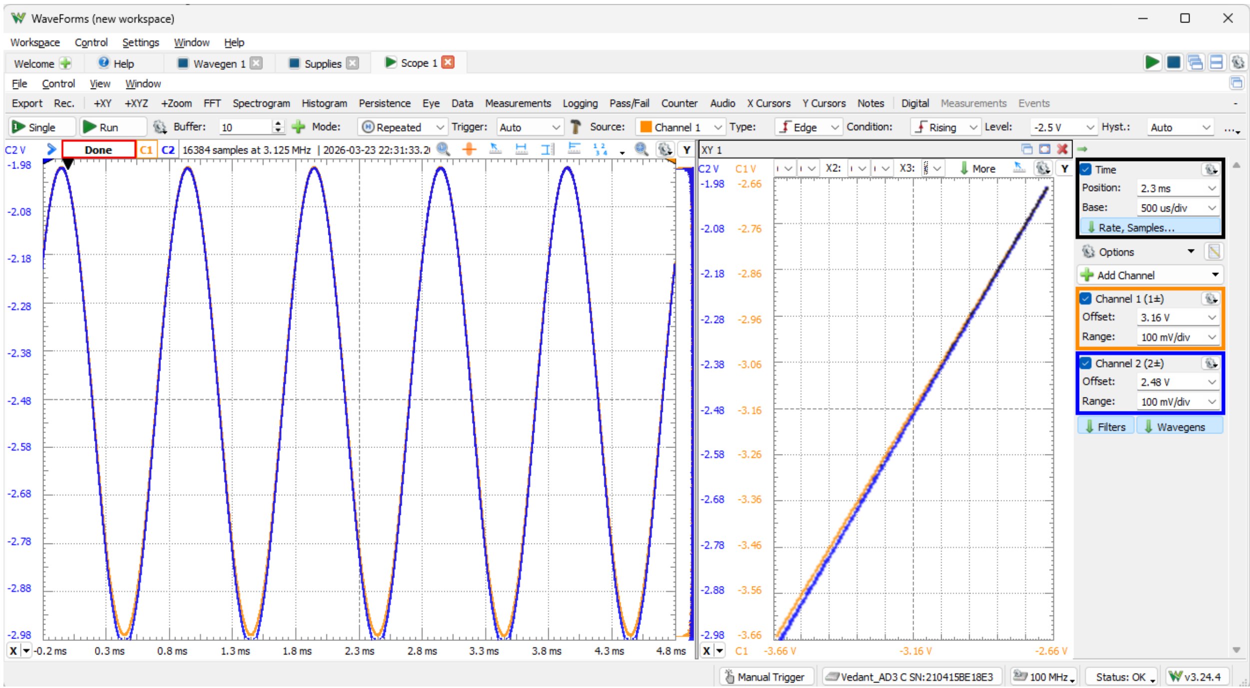


Figure 17: XY plot of output voltage vs input voltage demonstrating linearity with a slope of approximately 1.

Figure 10 shows the output signal vs the input signal using 2 plots, waveforms as functions of time and XY plot. It clearly demonstrates linearity.

6 Conclusion

The common-collector amplifier was designed to satisfy both the attenuation and signal swing requirements. Using the design targets

$$R_{in} \approx 10\,\mathrm{k\Omega}, \qquad r_e \approx 1\,\Omega$$

led to a desired emitter current of approximately

$$I_E \approx 25\,\mathrm{mA}$$

which then gave a desired base bias of approximately

$$V_B \approx -1.8\,\mathrm{V}$$

A first resistor choice of $R_1 = 2.2\,\mathrm{k\Omega}$ and $R_2 = 1\,\mathrm{k\Omega}$ was rejected because it made R_{in} too small. A final choice of

$$R_1 = 24.9\,\mathrm{k\Omega}, \qquad R_2 = 10\,\mathrm{k\Omega}$$

was selected instead. This design satisfies the required gain and keeps the transistor operating in the intended small-signal region.